

Bundibugyo virus, Democratic Republic of the Congo and Uganda, 2026

Latent Outbreak Visibility System (LOVS) applied to the 2026 BDBV outbreak. Ascertainment, detection depth, and pre-committed methodology calibration points, as of 20 May 2026.

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Bottom line. This is a **methodology contribution** in support of the DRC Ministry of Public Health, Hygiene and Social Welfare, the Uganda Ministry of Health, the World Health Organization, and Africa CDC. **It is not a forecast, a travel advisory, or a deployment recommendation.** The 20 May 2026 snapshot indicates the public reporting picture captures only an estimated **59 to 63 percent of laboratory-confirmable cases** and that detection occurred after multiple silent transmission generations, both intrinsic to early-stage filovirus surveillance. The corridor watchlist is a pre-committed calibration test of the method's uncertainty quality, not a ranked deployment list, and on historical data the method's corridor ranking does not beat a simple proximity or caseload baseline (see calibration section).

Why a methodology brief, today. As of May 20, 2026, the most prominent public quantitative output for this outbreak is the [joint WHO-Imperial College MRC GIDA report \(20 May 2026 update\)](#) estimating **400-900 total cases in DRC** (values over 1,000 not excluded), via population-movement extrapolation and deaths-back-projection through the case-fatality ratio. This brief treats that estimate as the academic reference for outbreak size, and reproduces its deaths-back-projection (Method 2, CFR 26/33/40, central 14-day doubling) to within a few cases. What this brief adds is the work the size estimate does not do: it is a convergence point that reconciles the scattered public sources into one provenance-tracked, source-conflict-aware view, and it publishes a reporting-completeness posterior, a pre-committed calibration set, and a cross-border corridor-risk view with date-stamped resolution. Within the archived source set for this snapshot, the other reviewed public outputs (WHO Disease Outbreak News 2026-DON602, the WHO AFRO Weekly Sitrep, Africa CDC PHECS declaration, and US CDC HAN 00530) do not include this combination. No comparable public output from the WHO Hub for Pandemic and Epidemic Intelligence in Berlin or the US CDC's Center for Forecasting and Outbreak Analytics was identified in this review as of the snapshot date. That gap is what this brief is built to fill. It complements the WHO-Imperial estimate; it does not replace it.

Authorities and standing. The Democratic Republic of the Congo (DRC) Ministry of Public Health, Hygiene and Social Welfare officially declared this outbreak on 15 May 2026 and is the lead authority on the DRC response, with the National Institute of Biomedical Research (INRB) confirming Bundibugyo virus (BDBV) by polymerase chain reaction (PCR). The Uganda Ministry of Health (MoH) is the lead authority on the Uganda response and confirmed two imported cases in Kampala on 15-16 May 2026. The World Health Organization (WHO) Director-General determined the outbreak a Public Health Emergency of International Concern (PHEIC) on 16 May 2026; WHO published the public statement on 17 May 2026. The Africa Centres for Disease Control and Prevention (Africa CDC), acting on the recommendation of the Emergency Consultative Group, declared the outbreak a Public Health Emergency of Continental Security (PHECS) on 18 May 2026. This brief is a methodology contribution in support of those authorities. It is not a substitute for them and does not speak on behalf of any of them.

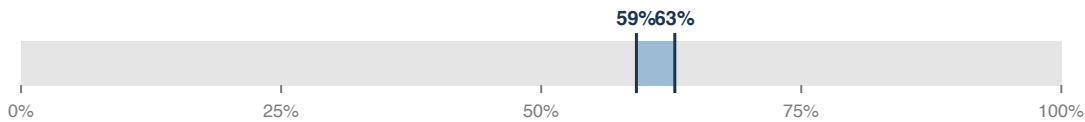
At a glance. Ebola disease caused by BDBV, a less common species of Ebola first identified in Uganda in 2007, is spreading in eastern DRC with cross-border and inter-province detection events. Public counts as of 20 May 2026: **53 laboratory-confirmed cases** (WHO Director-General remarks: 51 confirmed in DRC plus 2 in Kampala; earlier anchors are 10 confirmed as of 17 May per WHO PHEIC after Kinshasa deconfirmation, and 30 as of 19 May per ECDC), approximately **395 to 653 suspected cases** (Africa CDC PHECS 18 May to archived 20 May consensus source), and **106 to 144 deaths**, including four healthcare worker deaths at Mongbwalu General Referral Hospital within a four-day span per WHO DON 602. This brief applies the **Latent Outbreak Visibility System (LOVS)**, a stdlib-only Python pipeline that quantifies (a) the ascertainment gap, (b) the detection-depth posterior, and (c) inter-zone corridor risk under explicit calibration. The method was calibrated against the 2014 West Africa Ebola epidemic (a Zaire-species outbreak); applying it to a Bundibugyo-species outbreak introduces a species-transfer uncertainty that is reported honestly below.

What this brief is NOT. It is NOT a travel advisory. It is NOT a recommendation to restrict cross-border movement, close markets, or redirect commercial activity. It is NOT a deployment recommendation. The named corridors below are descriptive watch points for further investigation; they are NOT predictions of where the outbreak will spread. The corridor-risk numbers reflect a known model property (see methodology caveat) that overstates per-zone risk when confirmed cases are distributed across multiple health zones, as is the case here. Decisions on travel, trade, surveillance allocation, and response posture rest with the DRC Ministry of Public Health, the Uganda Ministry of Health, WHO, and Africa CDC. The pre-committed calibration points are not surveillance forecasts.

1. Ascertainment gap is wide and quantifiable

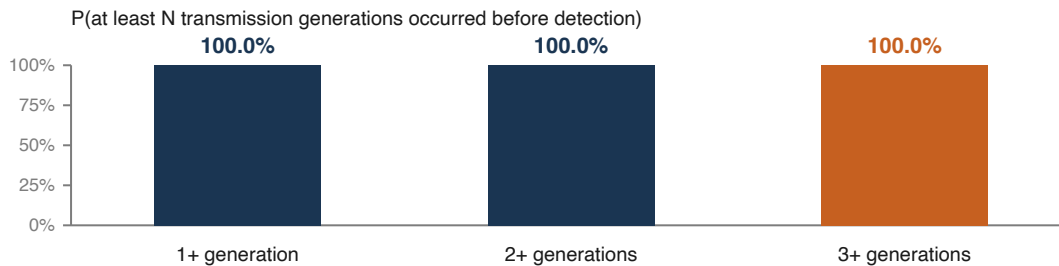
Reporting completeness (the share of underlying cases reflected in public counts), 50% uncertainty range: **[59.2%, 62.8%]**. This is consistent with what is typical of early-stage filovirus outbreaks under any surveillance system, where the gap between the visible surface and the underlying outbreak reflects the inherent reporting delay (Camacho 2015 PLOS Currents, an Ebola-Zaire onset-to-notification delay applied as a Bundibugyo proxy), the historical pattern of late Bundibugyo detection (Wamala 2010 EID), and operational realities of Ituri-region surveillance: ongoing insecurity in eastern DRC (per the Armed Conflict Location and Event Data (ACLED) project), internally displaced populations, and malaria plus other febrile, gastrointestinal, arboviral, or influenza-like illnesses that can complicate early clinical triage. The ascertainment gap is a structural feature of early outbreak reporting, not a critique of the national response.

50% uncertainty interval: 59.2% to 62.8% of cases visible in the public picture.



2. Detection occurred after multiple silent transmission rounds

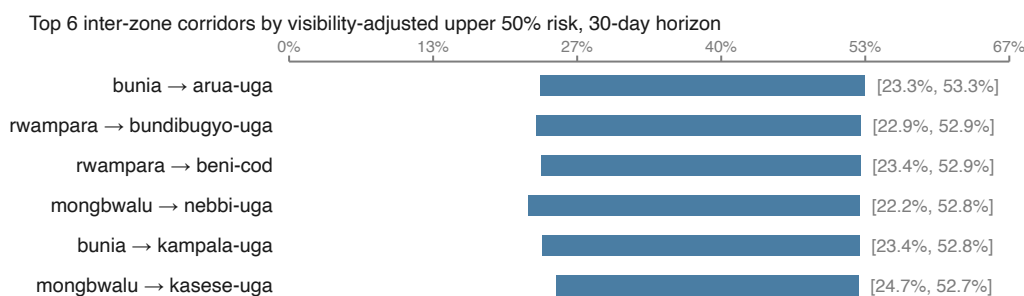
Posterior probability that at least three transmission generations (person-to-person rounds of spread) had already occurred before the outbreak became visible: **100.0%**. With 53 laboratory-confirmed cases, the branching-process inversion places near-unity mass on three or more silent generations preceding public detection under the current R and under-ascertainment priors. Wamala 2010 and MacNeil 2010 anchor the Bundibugyo-species incubation and interval context, but the numeric bins in this chart are driven by confirmed cases, the interim R prior, and the under-ascertainment prior. The R prior remains an explicitly labeled modeling assumption because a direct Bundibugyo-virus R0 estimate has not been located.



3. Corridor watch list (descriptive, not a ranking, not a forecast)

18 inter-zone corridors at a 30-day horizon. Across all 18, the upper bound of the ascertainment-adjusted 50% uncertainty range clusters tightly in the band 49.5 to 53.3 percent (the lower bound spans 20.9 to 24.7 percent). The tight clustering across all 18 corridors is itself the signal: no single corridor stands above the others; on historical data the method does not out-rank a simple proximity or caseload baseline (see calibration section).

Methodology caveat (load-bearing). The corridor model uses the snapshot's headline confirmed count (53 as of 20 May 2026) as a single aggregate input, then applies that aggregate to each named source zone. The official / regional record anchors the source-zone set and count trajectory (WHO PHEIC: 10 confirmed as of 17 May after Kinshasa deconfirmation; ECDC: 30 confirmed as of 19 May; WHO Director-General remarks: 53 confirmed as of 20 May). Because the 20 May official count is still not zone-attributed T0/T1 line-list data, and because cases are distributed across multiple health zones in Ituri Province plus North Kivu/Goma spillover detections, this simplification overstates per-zone risk. Read these numbers as an upper envelope until zone-attributed confirmed counts replace the aggregate input. Separately, the corridor model's gravity parameters (the population, road, healthcare-distance, and conflict exponents and the clamp) are transparent engineering heuristics, not fitted to a mobility dataset: Backer & Wallinga 2016 is the West Africa 2014 validation substrate and supports the broad gravity-type model family, but it does not source-fit the current LOVS constants (see data/evidence-chains.json, ec: lovs:module-d:corridor-gravity-exponents).



Read as a watch list, not a ranking decisive enough to direct deployment.

4. Four pre-committed methodology calibration points, resolving 19 June 2026

Four named corridors below are pre-committed on 20 May 2026 as calibration points for the underlying method. Each is paired with the model's ascertainment-adjusted 50% uncertainty range for a binary outcome at a 30-day horizon. On 19 June 2026, each outcome will be scored against publicly available reports from the DRC Ministry of Public Health, the Uganda Ministry of Health, the WHO, and Africa CDC, in order to evaluate whether the model's uncertainty ranges contained the observed outcome and to update the historical calibration. **These calibration points are NOT recommendations for the active public-health response.**

What each calibration point claims, in plain language:

Calibration point identifier	Plain-language statement
calibration-point:bdbv-uga-cod-2026:30d:e77003de63	At least one new laboratory-confirmed BDBV case appears in Kampala (Uganda) between 20 May 2026 and 19 June 2026, given continued reporting from Bunia Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: [22.9%, 52.3%] .
calibration-point:bdbv-uga-cod-2026:30d:bede8179ea	At least one new laboratory-confirmed BDBV case appears in Bundibugyo District (Uganda) between 20 May 2026 and 19 June 2026, given continued reporting from Rwampara Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: [22.7%, 52.3%] .
calibration-point:bdbv-uga-cod-2026:30d:d853267eed	At least one new laboratory-confirmed BDBV case appears in Beni Health Zone (North Kivu Province, DRC) between 20 May 2026 and 19 June 2026, given continued reporting from Mongbwalu Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: [21.8%, 52.2%] .
calibration-point:bdbv-uga-cod-2026:30d:cf92d9f82b	At least one new laboratory-confirmed BDBV case appears in Kasese District (Uganda) between 20 May 2026 and 19 June 2026, given continued reporting from Rwampara Health Zone (Ituri Province, DRC). Model's ascertainment-adjusted 50% uncertainty range: [20.9%, 51.5%] .



Historical calibration on the 2014 West Africa Ebola epidemic

The method has been tested retrospectively against the 2014 West Africa Ebola epidemic (Backer & Wallinga 2016 substrate, 62 prefectures across 74 weeks), where the eventual outcomes are public knowledge. Three runs are reported: without local context, with country-level local context (population density, road density, healthcare access, conflict intensity), and with district-level local context. Adding local context at country level tightens uncertainty (interval score down by roughly 35%); it does NOT improve discrimination of individual corridors above chance. The 2014 substrate was a Zaire-species outbreak; the method's calibration on Zaire data transfers to a Bundibugyo-species outbreak with a species-transfer uncertainty that is not separately quantified here.

Metric	No local context	Country-level	District-level	Change vs country
Brier score (probability accuracy; lower is better)	0.0586	0.0590	0.0590	+0.0000
Interval score (Bracher 2021; uncertainty quality; lower is better)	0.1002	0.0649	0.0649	+0.0000
Calibration error (predicted vs observed; lower is better)	0.0391	0.0500	0.0500	+0.0000

Finding. Country-level local context tightens uncertainty by approximately 35% but does not lift discrimination above chance, and district-level granularity does not add further. Candidate next-levers, ranked by expected impact: mobility data (Wesolowski 2016 Journal of Infectious Diseases), time-varying context (real-time political-violence intensity from ACLED and real-time MoH reporting cadence), re-engineering how the four context variables combine, and a richer transmission model. The calibration-error increase from 0.039 to 0.050 between no-context and context runs is small and most likely an artifact of the small evaluation set (five as-of dates).

Robustness check (skill, discrimination, calibration). A rolling-origin backtest across a pre-registered grid of as-of weeks, reproducible via `robustness_backtest.py`, refines the table above. At the early checkpoints the method ranks which prefectures appear next with a ROC AUC of 0.72, above the 0.50 chance level, but a distance-only baseline (0.73) and a source-load-only baseline (0.72) rank just as well: the gravity and covariate machinery add no ranking value, so the discrimination that exists is the spatial autocorrelation of the epidemic rather than the model. Against a base-rate (climatology) reference the Brier skill score is -0.03 (95 percent interval -0.48 to 0.09); across the full grid no configuration shows a skill interval above zero at any window, so the method has no positive calibration skill beyond predicting the base rate. Intervals are from a target-prefecture clustered bootstrap. These are 2014 West Africa (Zaire-species) results and are not a portable skill claim for a Bundibugyo-species outbreak.

If you have point-of-care data

This brief reports method estimates from publicly aggregated reporting only. If you are working directly in the affected zones, you almost certainly hold information that is privileged, time-sensitive, and not appropriate for a public repository. **Please do not paste line-list rows, GPS-tagged case locations, sequencing reads, or any identifying detail into a public GitHub issue.** You can reach me directly at frans@arcede.com if any of the following would help your work.

- **Onset-date line-list extract (de-identified).** Even a partial onset-date histogram for one health zone substantially narrows the latent-active-chains plausibility interval emitted by Module D.
- **Zone-attributed case counts.** The load-bearing simplification flagged in Panel 3 is that the full confirmed count is contributed to each named source zone. A mapping of `{zone_id: count}` for the affected districts is the largest single discrimination lever the method is missing.
- **Validated zone GPS centroids.** The published map ships verified centroids for zones currently in scope (`data/zones.json`). For zones we may have missed, a centroid plus a one-sentence rationale is enough to extend the corridor model.

- **Mobility traces or transport-flow snapshots.** Call-detail-record summaries or surveyed transport flows, even at admin-2 aggregation, are the documented next-lever for moving the method above chance discrimination (Wesolowski 2015 PNAS).
- **Case-confirmation latency.** Field-observed delays (sample collection to PCR result, in days) are a direct prior update for Module C, replacing the assumed 2014 West-Africa delay distribution.

Contributions that land in the repository are cited and timestamped. If you prefer to remain unnamed, I can credit you by initials or pseudonym at the time of contribution.

Known blindspots and calibration design notes

- **Mahagi as a source zone is still outside the model, and the Arua/Nebbi corridors are on the watchlist but not yet pinned.** Mahagi (DRC) and Goli (Uganda) form one of DRC's busiest land border crossings on the East African Northern Corridor, with 95,000+ refugees at the Rhino Camp settlement near Arua (UNHCR, late-2025). The 21 May 2026 revision added Arua District and Nebbi District to the candidate-target watch set, so the snapshot now carries corridors into both. They sit in the descriptive watchlist, not in the four pre-committed calibration points: the 20 May calibration set is carried forward unchanged so the 19 June scoring contract holds, and new corridors can only be pinned as a new date-stamped block. Mahagi (DRC) is still not modeled as a source zone, because no WHO or Africa CDC source names Mahagi health zone as case-affected; `snapshot_sensitivity.py` shows where the omitted Mahagi-to-Arua corridor would rank under an explicit equal-burden counterfactual.
- **The four pre-committed calibration corridors cluster within a narrow probability band (20.9 to 22.9% lower, 51.5 to 52.3% upper).** A calibration set that spans a wider band, including at least one corridor with a model probability in the 5 to 20% range, would be a stronger discrimination test.
- **The corridor mongbwalu-to-beni-cod may be confounded.** Beni Health Zone sits in North Kivu Province, which the 18 May Africa CDC reporting identifies as already part of the outbreak footprint. A positive resolution on this corridor may simply reflect Beni being part of the active outbreak rather than source-zone-to-target-zone transmission.
- **Conflict-state coverage is qualitative.** The brief invokes CODECO and ADF activity in Ituri/North Kivu and 7.3 million IDPs in eastern DRC as descriptive context, but the per-zone conflict-intensity input is the 2014 West Africa ACLED snapshot used in the Mode A historical calibration. A 2026 ACLED snapshot for eastern DRC is a documented next-lever.

What this brief does NOT claim

- **Not a forecast.** The four pre-committed calibration points exist to evaluate the method's uncertainty quality at resolution. They are not surveillance recommendations and not deployment recommendations.
- **Not a critique of the national response.** Ascertainment gaps and late detection of filoviruses are intrinsic to the pathogen and to the operational context (security, displacement, co-circulating pathogens), not to the speed of any specific national response. The DRC Ministry of Public Health declared on 15 May, the Uganda Ministry of Health confirmed imported cases on 15-16 May, INRB confirmed BDBV by PCR within days. This brief takes the national declarations as the authoritative timeline.
- **Does not replace field epidemiology.** Line-listing, contact tracing, genomic sequencing, and clinical reasoning are where outbreak control happens.
- **Species-transfer uncertainty is not separately quantified.** The historical calibration substrate is a Zaire-species outbreak; transfer to Bundibugyo carries unquantified uncertainty in the priors and the corridor model.
- **Numbers are prior-dominated at this case count.** The ascertainment and transmission-depth posteriors are heavily informed by the prior delay and transmission distributions, not by data alone. In particular, the branching-process reproduction prior is an interim modeling assumption: no Bundibugyo-virus-specific basic reproduction number has been estimated in the published literature (Van Kerkhove 2015), so the detection-depth result is prior-driven and species-transferred, not a data-driven BDBV estimate. Sensitivity analyses across alternative priors are a recommended next step.
- **Open to independent replication.** Code is Apache 2.0 licensed; original authored methodology, prose, schema, and derived artifacts are CC BY 4.0; third-party source material and extracted publisher-owned tables retain their original terms. The method is described in CITATIONS.md. Independent replication is welcomed.

Citation. Moore F. *Bundibugyo virus, DRC and Uganda, 2026: surveillance methodology brief*. Released 2026-05. <https://github.com/ArcedeDev/bdbv-2026-lovs>. | **Live data:** WHO Disease Outbreak News item 2026-DON602, WHO African Region Weekly External Situation Report 01 (data as of 18 May 2026), Africa CDC PHECS declaration (18 May 2026), and aggregated public reporting through 20 May 2026. | **Methodology citations:** Wamala 2010 Emerging Infectious Diseases (EID), MacNeil 2010 EID, Albariño 2013 Virology, Faye 2015 Lancet ID, Backer & Wallinga 2016 PLOS Computational Biology, Bracher 2021 PLOS Computational Biology (full list at CITATIONS.md). | **License.** Code Apache 2.0; original authored methodology/prose/schema/derived artifacts CC BY 4.0; third-party source material keeps publisher terms; WHO content Creative Commons BY-NC-SA 3.0 IGO. | **Pre-commitment timestamp:** verifiable via the git commit history of `data/live-bdbv-2026-output.json` in the source repository (first committed on 2026-05-20, before the 2026-06-19 resolution window). | **Resolution:** 2026-06-19T23:59:59Z.